

1 The Role of Statistics in Engineering

We start with two examples from the bachelor **Military Systems and Technology**.

- **Maintenance: the reliability of a circuit**

A circuit operates only if there is a path of functional devices from left to right. Devices *A* and *B* are connected in parallel, device *C* is in series.

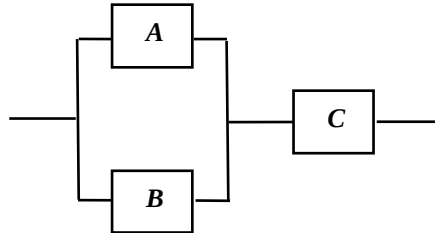


Figure 1.1 Circuit with three devices

For example, the circuit operates if the devices *A* and *C* function, but the circuit fails, if device *C* does not function.

Assume that all devices function on average for 10 hours. What is the probability that the circuit will operate for at least 9 hours?

We look at this problem in two different ways, first **deterministic** (i.e. there is no variability), then **stochastic** (i.e. there is variability).

- **deterministic**

All devices will function for precisely 10 hours. The requested probability is therefore equal to 1.

- **stochastic**

Both a **simulation** using **Excel** and a **calculation** (later in this course) result in a probability of approximately 0.26.

See the Excel-file “PMTPA_STAT_1_stochastic.xlsx” or the print screen in Figure 1.2.

In this file, the above circuit is simulated 10,000 times. In each row, three life lengths for the devices *A*, *B* and *C* are generated. The life length of the circuit is calculated in column “D”. Next, in column “E”, it is determined whether the circuit still functions after 9 hours. In that case, the number 1 is shown in column “E”.

The requested probability is then calculated in cell “E10003”. The magnitude of this probability is partly dependent on the number of simulations. Use the button “F9” to examine the sensitivity at this point.

PMTPS_STAT_1_stochastic.xlsx - Microsoft Excel

	A	B	C	D	E	F	G	H	I
1	Device A	Device B	Device C	Circuit					
9988	6,841	14,873	8,209	8,209	0				
9989	4,240	8,650	15,175	8,650	0				
9990	18,044	6,365	38,085	18,044	1				
9991	1,194	1,557	5,246	1,557	0				
9992	38,282	8,686	0,611	0,611	0				
9993	7,397	14,077	18,563	14,077	1				
9994	4,863	1,553	14,752	4,863	0				
9995	6,554	5,390	2,647	2,647	0				
9996	6,162	32,267	10,679	10,679	1				
9997	3,081	1,041	11,934	3,081	0				
9998	15,068	5,047	97,078	15,068	1				
9999	3,645	0,658	5,429	3,645	0				
10000	9,057	14,896	0,006	0,006	0				
10001	3,125	17,618	7,873	7,873	0				
10002	Mean A	Mean B	Mean C	Mean Circuit	Probability				
10003	9,983	10,122	9,876	6,644	0,2638				

Figure 1.2 Life length of a circuit

- Operations Research: Search and Detection

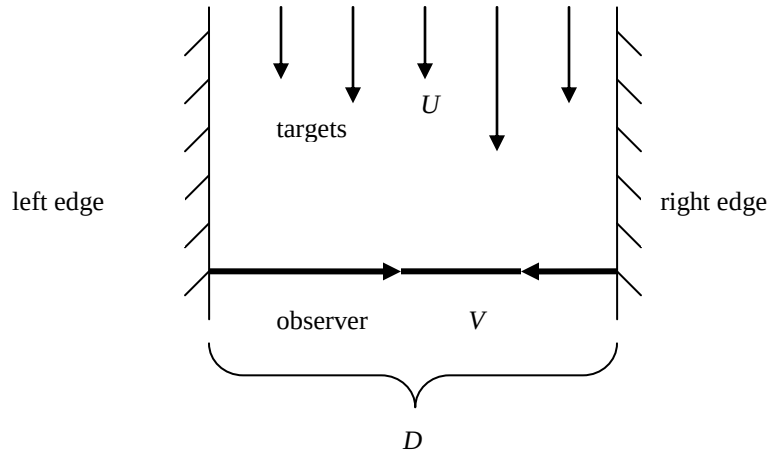


Figure 1.3 Linear barrier (reversing course at the edge)

An **observer** is protecting a lane with width D while moving at speed V through the lane according some fixed pattern. Any target that closes the observer to within his sweep radius R is detected. So the observer's detecting device is binary: the target is either detected or not detected.

We assume that **targets** intend to traverse this lane Southwards. We also assume that the target speed U is constant along its path. A further assumption is that we know the intent and the capabilities of the target. More precisely, we assume that we know its speed. Its position,

however, is unknown, so arbitrary. In stochastic terms: all positions have an equal probability of occurrence.

Since the target knows that a **preference for the edges** (i.e. positions nearby the edges of the lane are more likely than positions in the centre of the lane) leads to a lower probability of detection, the target chooses its position based on the probability function in Figure 1.4.

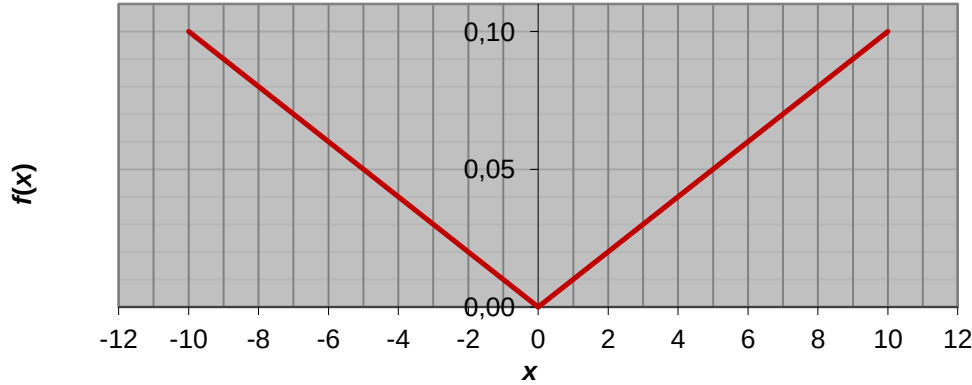


Figure 1.4 Probability function

So if $D = 20$ nm, the probability that the target chooses a position between -5 nm and $+5$ nm is equal to 0.25 (i.e. the area under the graph of the probability function between $x = -5$ and $x = +5$), while the probability that the target chooses a position between -10 nm and -5 nm or between 5 nm and 10 nm is equal to .75.

What is the probability that the target will be detected?

Using probability, calculus and geometry, it is not difficult to derive the following formula

$$P_{\text{det}} = \left\{ \frac{4R^3}{3D^3} \cdot \left(1 + \left(\frac{V}{U} \right)^2 \right) - \frac{2R^2}{D^2} \cdot \sqrt{1 + \left(\frac{V}{U} \right)^2} + \frac{2R}{D} \right\} \cdot \sqrt{1 + \left(\frac{V}{U} \right)^2}.$$

Example 1

$$\left. \begin{array}{l} D = 20 \\ R = 4 \\ V = 18 \\ U = 12 \end{array} \right\} \Rightarrow P_{\text{det}} = 0.5236$$

2.1 Sample Spaces and Events

E = taking a target under fire until the target is eliminated

$S = \{ h, mh, mmh, \dots \}$, where h = hit and m = miss

A = "hit shooting in no more than 3 attempts" = $\{ h, mh, mmh \} \subset S$

B = "hit shooting in at least 2 attempts" = $\{ mh, mmh, mmmh, \dots \}$

Random Experiment
Sample Space
Event

Remarks

- $A' =$ complement of A = not- A
- $A \cap B$ = intersection of A and B = both A and B
- $A \cup B$ = union of A and B = A (but not B) or B (but not A) or both A and B
- if $A \cap B = \emptyset$, then A and B are mutually exclusive

Example 1 (see above data)

- $B' = \{ h \}$ "hit in one go"
- $A \cap B = \{ mh, mmh \}$ "hit in 2 or 3 attempts"
- $A \cup B = S$ "certain event"

2.2 Axioms of Probability

Let S be a **sample space** and E any **event** in a **random experiment** E .

A function

$$P: E \subset S \rightarrow \mathbb{R}$$

satisfying

(A.1) $P(S) = 1$

Certain Event

(A.2) $0 \leq P(E) \leq 1$, if $E \subset S$

(A.3) if $E_1 \cap E_2 = \emptyset$, then $P(E_1 \cup E_2) = P(E_1) + P(E_2)$

Mutually Exclusive

is called a **probability function**.

2.3 Probability Rules

R.1 $P(\emptyset) = 0$

Impossible Event

R.2 $P(A') = 1 - P(A)$

Complement Rule

R.3.1 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

General Addition Rule

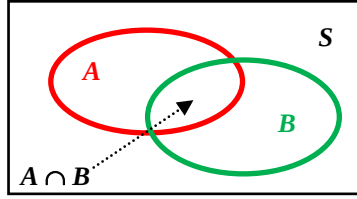


Figure 2.1 The union of the events A and B

Proof (only R.3.1)

$$A \cup B = A \cup (B \cap A') \quad \Rightarrow \quad P(A \cup B) = P(A) + P(B \cap A') \quad \text{A.3}$$

$$B = (A \cap B) \cup (B \cap A') \quad \Rightarrow \quad P(B) = P(A \cap B) + P(B \cap A') \quad \text{A.3}$$

$$P(A \cup B) - P(B) = P(A) - P(A \cap B) \quad \Leftrightarrow$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{R.3.2} \quad P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

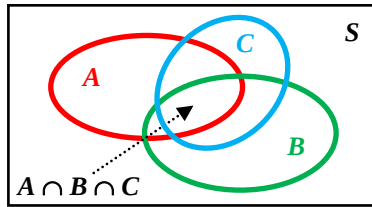


Figure 2.2 The union of the events A, B and C

Example 2

E = tossing with two fair dice

$S = \{ (1,1), (1,2), \dots, (6,5), (6,6) \}$

A = "sum of the two dice is at least 10"

B = "sum of the two dice is no more than 10"

Determine $P(A)$, $P(B)$, $P(A \cap B)$ and $P(A \cup B)$.

Solution

$$P(A) = P(\text{sum is 10 or 11 or 12}) = P(\{(4,6), (5,5), (5,6), (6,4), (6,5), (6,6)\}) = \frac{6}{36} = \frac{1}{6}$$

$$P(B) = 1 - P(B') = 1 - P(\text{sum 11 or 12}) = 1 - P(\{(5,6), (6,5), (6,6)\}) = 1 - \frac{3}{36} = \frac{11}{12}$$

$$P(A \cap B) = P(\text{sum is 10}) = P(\{(4,6), (5,5), (6,4)\}) = \frac{3}{36} = \frac{1}{12}$$

$$P(A \cup B) = \begin{cases} P(S) = 1 \\ P(A) + P(B) - P(A \cap B) = \frac{1}{6} + \frac{11}{12} - \frac{1}{12} = 1 \end{cases}$$

2.4 Conditional Probability

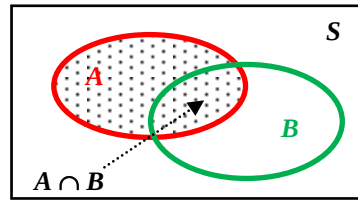


Figure 2.3 Conditional probability of B given A

The **conditional probability** of an event B given an event A , denoted as $P(B|A)$, is

$$P(B|A) = \frac{P(A \cap B)}{P(A)}.$$

2.5 Multiplication and Total Probability Rules

R.4.1 $P(A \cap B) = P(A) \cdot P(B|A)$

Multiplication Rule

R.4.2 $P(A \cap B) = P(A|B) \cdot P(B)$

Multiplication Rule

R.5 $P(B) = P(B|A) \cdot P(A) + P(B|A') \cdot P(A')$

Total Probability Rule

Proof (only R.5)

$$\begin{aligned} B &= (B \cap A) \cup (B \cap A') & \Rightarrow & & P(B) &= & P(B \cap A) + P(B \cap A') & \text{A.3} \\ & & & & &= & P(B|A) \cdot P(A) + P(B|A') \cdot P(A') & \text{R.4.2} \end{aligned}$$

Remark

A **tree diagram** can be used to display conditional probabilities. See the example below.

Example 3 (Example 2-27 in the textbook)

Suppose that in semiconductor manufacturing, the probability is 0.10 that a chip subjected to high levels of contamination during manufacturing causes a product failure. The probability is 0.005 that a chip not subjected to high contamination levels during manufacturing causes a product failure. In a particular production run, 20% of the chips are subject to high levels of contamination. What is the probability that a product using one of these chips fails?

Solution

Let A = chip is subjected to high levels of contamination. So $P(A) = 0.2$.

Let B = semiconductor fails. So $P(B|A) = 0.1$ and $P(B|A') = 0.005$. $P(B) = ?$

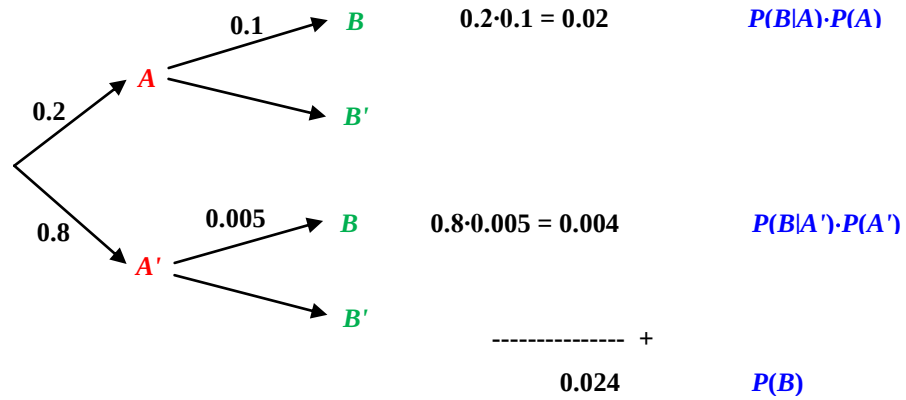


Figure 2.4 Tree diagram semiconductor manufacturing

Remark

In the general case, the **Total Probability Rule** is as follows:

Assume $S = \bigcup_{i=1}^k A_i$, and $A_i \cap A_j = \emptyset$ ($\forall i, j$), then

$$P(B) = \sum_{i=1}^k P(B|A_i) \cdot P(A_i).$$